

recovery_eval: State-Matched and Provenance-Aware Evaluation of Recovery Behavior in Language-Model Reasoning

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Abstract—Evaluating whether different language-model policy configurations exhibit improved recovery from intermediate reasoning errors is difficult because overall continuation quality can confound recovery-specific behavior. We introduce `recovery_eval`, a reproducible data-centric evaluation framework that compares verifier-defined recovery states with prospectively matched reference-control states using frozen structural covariates. The framework combines state matching with append-only exposure governance, primitive neural-rollout provenance, and independent analysis reconstruction. We apply `recovery_eval` to 400 genuine neural continuations generated by two released Qwen2.5-Math 1.5B checkpoint-interface configurations on 20 prospectively isolated GSM8K evaluation problems. The Instruct checkpoint showed higher observed continuation success than the Base checkpoint in both recovery and matched-control states, but the improvement was larger for controls. The resulting matched recovery-specific contrast was minus 0.110, with a descriptive 95 percent problem-level bootstrap interval from minus 0.240 to 0.030. Thus, under this evaluation, we did not observe evidence of a recovery-specific advantage for the Instruct checkpoint. The result illustrates how aggregate benchmark improvements can obscure state-specific reasoning behavior and motivates provenance-aware, state-matched evaluation for language-model diagnostics.

Index Terms—Large Language Models, Mathematical Reasoning, Model Evaluation, Data-Centric AI, Reproducibility

I. Introduction

Large language models (LLMs) trained on multi-step mathematical reasoning demonstrate remarkable capacity for generating complex chain-of-thought solutions [1], [2]. A foundational question in AI evaluation is whether different released language-model configurations exhibit recovery behavior beyond their general continuation-quality differences [3], [4].

Evaluating error-recovery capability is fraught with statistical confounding. Standard benchmark evaluation measures aggregate end-to-end accuracy, which conflates baseline generation fluency with true error recovery. Simply comparing model rollouts from error-containing prefixes against valid prefixes ignores critical confounding covariates, including trajectory depth, remaining solution complexity, problem difficulty, and token length. A post-trained model may exhibit higher accuracy on error prefixes simply because its overall baseline generation

quality has improved across all prompt conditions, rather than possessing a specialized error-recovery mechanism.

To address these diagnostic challenges, we introduce `recovery_eval`, a data-centric evaluation infrastructure for measuring state-specific recovery behavior in reasoning LLMs. The primary contributions of this work are three-fold:

- 1) **Verifier-Defined State Matching Protocol:** A prospective matching methodology that pairs verifier-identified error states with valid reference-control states using frozen structural covariates and an exact matching constraint on categorical problem attributes.
- 2) **Provenance and Exposure Governance Architecture:** An append-only evidence ledger that locks evaluation items, logs primitive BPE token-level generation continuations, verifies model weight manifests, and enables 100% independent analytical reconstruction.
- 3) **Genuine Two-Checkpoint Framework Demonstration:** A 400-rollout empirical evaluation across Qwen2.5-Math-1.5B Base and Instruct checkpoints, demonstrating that aggregate continuation success gains (+0.4300 on recovery vs +0.5400 on control) do not automatically translate into a detectable recovery-specific advantage ($D_{\text{recovery}} = -0.1100$).

II. Related Work

Recent research has focused on reasoning self-correction, test-time compute scaling, and process reward modeling [5], [6]. Table I compares `recovery_eval` with existing evaluation methodologies.

Existing benchmark designs generally do not explicitly control for the structural state differences targeted by our matched recovery/control protocol.

III. Problem Formulation

Let s represent a partial reasoning trajectory prefix for a mathematical word problem $p \in \mathcal{P}$. A deterministic verifier $\mathcal{V}(s) \in \{0, 1\}$ evaluates whether prefix s contains any logical or arithmetic errors.

A recovery state s_R is a partial prefix containing a verifier-identified intermediate error. A reference-control

TABLE I
Comparison with Existing Reasoning Evaluation Frameworks

Framework Work	State Matching	Exposure Ledger	Primitive Provenance
End-to-End GSM8K [1]	None	No	Final Answer Only
PRM800K [3]	Unmatched	Partial	Step Scores
STaR [4]	Unmatched	No	Filtered Traces
recovery_eval (Ours)	Prospectively Matched	Append-Only LEDGER	Full BPE Token + Raw JSONL

state s_C is a valid intermediate prefix for the same problem instance at an equivalent stage of resolution.

For a target policy π and baseline policy π_0 , let $V(s) \in \{0, 1\}$ denote the binary continuation success indicator of completing the solution correctly from prefix s . We define the matched recovery-specific contrast D_{recovery} as:

$$D_{\text{recovery}} = \mathbb{E}[V_\pi(s_R) - V_{\pi_0}(s_R)] - \mathbb{E}[V_\pi(s_C) - V_{\pi_0}(s_C)] \quad (1)$$

IV. The recovery_eval Framework

The recovery_eval framework provides an end-to-end evaluation pipeline comprising three main modules: state generation, prospective matching, and governance-backed neural rollout execution.

A. System Architecture

The framework operates as a decoupled pipeline: 1) Registry Phase: Evaluation items and candidate prefixes are locked in an append-only registry; 2) Matching Phase: Recovery states are prospectively matched to reference control states prior to inference; 3) Execution Phase: Continuations are sampled under sealed generation settings and evaluated by the primitive verifier.

V. State Construction and Prospective Matching

Recovery states s_R are constructed by introducing controlled single-step arithmetic perturbations into valid intermediate reasoning prefixes. Reference control states s_C maintain valid prefix steps.

To prevent confounding due to trajectory length or problem complexity, each recovery state s_R is matched to a control state s_C using a normalized weighted-L1 Manhattan distance over continuous structural covariates:

$$d(i, j) = \sum_{k=1}^K w_k \frac{|x_{ik} - x_{jk}|}{s_k} \quad (2)$$

subject to an exact categorical matching constraint on reasoning operation type and problem difficulty. Table II lists the prospective matching parameters.

TABLE II
Frozen Prospective Matching Covariates and Scales

Covariate Name	Type	Weight (w_k)	Scale (s_k)
Trajectory Depth	Continuous	0.4	1.5
Remaining Solution Length	Continuous	0.4	1.0
Token Length	Continuous	0.2	15.0
Reasoning Operation	Categorical	Exact Match	–
Problem Difficulty	Categorical	Exact Match	–

VI. Provenance and Exposure Governance

To guarantee artifact integrity and prevent item re-selection or data leakage, recovery_eval implements an append-only event ledger. Every item exposure event is logged with cryptographic parent hashing [7], [8].

Each raw rollout record in RAW_NEURAL_ROLLOUTS.jsonl captures input token IDs, generated BPE token continuation IDs, monotonic generation timestamps, resolved HuggingFace commit hashes, weight manifest SHA-256 digests, and deterministic verifier outputs.

VII. Experimental Design

We evaluate the framework across 20 fresh, prospectively isolated GSM8K test items ($N = 20$). For each problem item, 1 recovery state and 1 matched control state are evaluated across 2 policy configurations (Qwen2.5-Math-1.5B Base and Instruct) using 5 stochastic rollout seeds ($S = 5$), yielding exactly 400 continuations ($20 \times 2 \times 2 \times 5 = 400$).

Table III details the execution design and provenance metrics.

TABLE III
Evaluation Design and Provenance Statistics

Attribute	Specification / Value
Base Model	Qwen/Qwen2.5-Math-1.5B (4a83ca6e)
Instruct Model	Qwen/Qwen2.5-Math-1.5B-Instruct (aafeb0fc)
Hardware Device	Apple Silicon MPS (mps:0)
Total Rollouts	400 (200 Base, 200 Instruct)
Total Generated Tokens	19,212 BPE tokens
Measured Duration	1,755.86 seconds (≈ 29.26 minutes)
Base Throughput	11.86 tokens/sec (11,354 tokens in 957.37s)
Instruct Throughput	9.84 tokens/sec (7,858 tokens in 798.49s)
BPE Decode Match	100.0% (400/400 exact match)
Raw JSONL SHA-256	51b5a157d9e44102caeb86d0b356f558...

VIII. Results

Table IV presents continuation success rates across states and policy configurations.

Under the evaluated state-matched protocol, we did not observe evidence of a recovery-specific advantage for the Instruct checkpoint over the Base checkpoint. The estimated matched recovery-specific checkpoint-interface

TABLE IV
Observed Empirical Continuation Outcomes and Contrast

State Condition	Base (π_0)	Instruct (π)	Difference (Δ)
Recovery States (s_R)	0.1500	0.5800	+0.4300
Control States (s_C)	0.3800	0.9200	+0.5400
Matched Contrast (D_{recovery})	–	–	–0.1100

contrast was -0.1100 , with a 95% descriptive problem-level bootstrap interval of $[-0.240, +0.030]$ (10,000 resamples).

The Instruct checkpoint showed higher continuation success than Base in both recovery (+0.4300) and control (+0.5400) states, but the improvement was larger for controls. Consequently, aggregate benchmark gains do not automatically translate into a recovery-specific advantage.

IX. Sensitivity and Reproducibility Analysis

We assess matching quality across the 20 matched pairs. The mean normalized weighted-L1 distance is $d_{\text{mean}} = 0.0360$, median is $d_{\text{median}} = 0.0360$, and maximum distance is $d_{\text{max}} = 0.0360$. All 20 pairs (20/20) satisfy both the standard ($d \leq 0.25$) and tight ($d \leq 0.10$) thresholds [9].

Table V reports covariate balance metrics. Standardized Mean Differences (SMDs) for trajectory depth and remaining length are +0.0000, while token length SMD is +0.1333.

TABLE V
Matching Distance, Covariate Balance & Threshold Sensitivity

Metric / Covariate	Raw Difference	SMD / Distance
Trajectory Depth	+0.0000	+0.0000
Remaining Length	+0.0000	+0.0000
Token Length	+2.0000	+0.1333
Mean Pair Distance (d_{mean})	–	0.0360
Max Pair Distance (d_{max})	–	0.0360

Independent re-computation from sealed raw evidence RAW_NEURAL_ROLLOUTS.jsonl (SHA-256: 51b5a157d9e4...) yields exact agreement (0.0 discrepancy across all metrics).

X. Limitations

This study has explicit methodological scope boundaries:

- **Model Scope:** Evaluated on a single model family (Qwen2.5-Math-1.5B) across two released checkpoints.
- **Benchmark Scope:** Evaluation uses 20 GSM8K test items ($N = 20$). Benchmark pretraining contamination cannot be excluded.
- **Sample Size:** 5 stochastic rollout continuations per state/policy represent evaluation samples, not independent model training replications.

- **No Causal Claim:** Results represent a descriptive contrast under state matching, not a causal claim regarding post-training mechanisms.

XI. Conclusion

We presented recovery_eval, a prospective state-matched evaluation framework for language-model reasoning diagnostics. By pairing verifier-defined recovery states with structural reference controls and maintaining primitive rollout governance, the framework enables fine-grained evaluation of reasoning policies without confounding baseline fluency gains.

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